Bank Marketing Campaign Performance Analysis

An end-to-end data analytics project using Python, MySQL, Power BI, and DAX to understand subscription behavior and support better marketing decisions.

Python — prepare

MySQL — analyze

Power BI — decide

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THE BUSINESS CHALLENGE

41,176 customers. 11.27% subscribed.

How can customer and campaign data support better targeting for future campaigns?

41,176

customers analyzed

4,639

subscribers

11.27%

overall subscription rate

THE QUESTION IS HOW CUSTOMER CHARACTERISTICS AND CAMPAIGN ACTIVITY RELATE TO SUBSCRIPTION DECISIONS.

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THE ANALYTICS WORKFLOW

From raw data to marketing decisions

01

Raw dataset

02

Python

prepare and structure

03

MySQL

analyze performance

04

Power BI

build KPIs and dashboards

05

Insights

recommendations

06

Documentation

GitHub

Each tool moves the project closer to an actionable business decision.

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01 / DATA PREPARATION

Preparing the data for analysis

The goal was a cleaner, analysis-ready dataset for SQL analysis and Power BI reporting.

01 Reviewed structure and data types

02 Checked missing and unknown values

03 Identified duplicate records

04 Created customer age groups

05 Created a numeric subscription indicator

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SQL turned the dataset into business questions



Overall performance



Job and age groups



Contact method



Education and marital status



Previous campaign results



Contact frequency

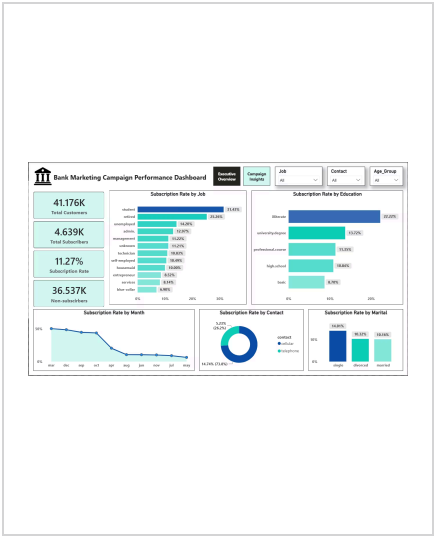
MySQL connected customer attributes and campaign activity to measurable subscription outcomes.

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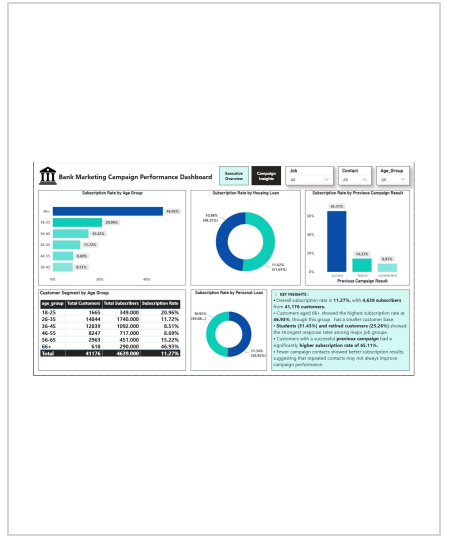
Power BI made the analysis decision-ready

Executive Overview: customers, subscribers, subscription rate, non-subscribers, performance by job, education, month, contact method, marital status.

Campaign Insights: age groups, housing/personal loans, previous campaign performance, customer segments, key insights.



EXECUTIVE OVERVIEW



CUSTOMER AND CAMPAIGN INSIGHTS

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WHAT THE DATA SHOWED

Previous success was the strongest signal

65.11%

subscription rate after a successful previous campaign

46.93%

subscription rate among customers aged 66+

Students and retired customers showed stronger response rates among major job groups.

Lower contact frequency was associated with better subscription results.

These patterns describe the available data; they do not prove cause and effect.

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FROM INSIGHT TO ACTION

Turn the patterns into smarter campaign choices

01

Improve targeting

Focus more attention on groups with stronger response rates.

02

Use campaign history

Incorporate previous outcomes into targeting decisions.

03

Review repeated contacts

Test more targeted follow-up strategies.

04

Segment customers

Tailor campaign approaches to different customer groups.

Project outcome: a complete workflow from Python preparation and SQL analysis to DAX KPIs, Power BI reporting, and documented recommendations.

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